Client
POST /predict
(audio file)

API layer [api.py]

FastAPI endpoint
check CNN
checkpoint is
loaded

Prediction [predict.py]

predict(audio_bytes)
calls cnn.run()

CNN Model [models/cnn.py + preprocess.py]

Preprocessing
audio load -> log-
mel spectrogram
-> normalize ->
tensor

Feature Extraction
4x
Conv/BatchNorm/ReLU
blocks
-> AdaptiveAvgPool2d

Classification
Linear -> ReLU ->
Linear
-> raw logits

Postprocessing [predict.py + labels.py]

Softmax -> per-
class scores

Map scores to
label names
highest score ->
label

Response to
client
{ label, probs,
filename }